UNIVERSAL ANIMAL WELFARE & AI ECOSYSTEM

# EcoTrack AI Platform Complete Architecture & Engineering Guide

## Dual-Engine Cloud/Offline Architecture, Python AI Microservice, Admin Governance & Codebase Reference

This official architectural guide provides an exhaustive engineering reference for the EcoTrack platform. EcoTrack is a multi-tier, offline-first animal welfare, species preservation, and real-time computer vision biomechanical diagnostic ecosystem. This document details the interaction flows between the Expo React Native Mobile Frontend, the Vanilla JavaScript Web Application, the Node.js Express Coordination Server, and the high-accuracy Python AI Microservice (Flask). Furthermore, it provides technical specifications on the dual-mode Supabase Cloud HTTPS / SQLite WAL database synchronization engine, the comprehensive Administrator Portal & Governance framework, biochemical scan analysis, universal messaging, and biological datasets.

**Project: EcoTrack Animal Welfare Ecosystem**
Edition: Version 5.0 Production Architecture Build
Status: Cloud Synchronized & Offline Dual-Engine Certified
Target Platforms: Web (Desktop/Mobile Browsers), Android, iOS
Release Date: November 2026
Generated by: Antigravity AI Engineering Suite

# 1. 4-Tier System Architecture & Topology

EcoTrack operates as an offline-first, high-throughput 4-tier hybrid ecosystem engineered to track animal rehabilitation, record environmental impact milestones, and perform edge/cloud computer vision diagnostics. The system decouples high-speed presentation and client logic from heavy deep-learning model inference:

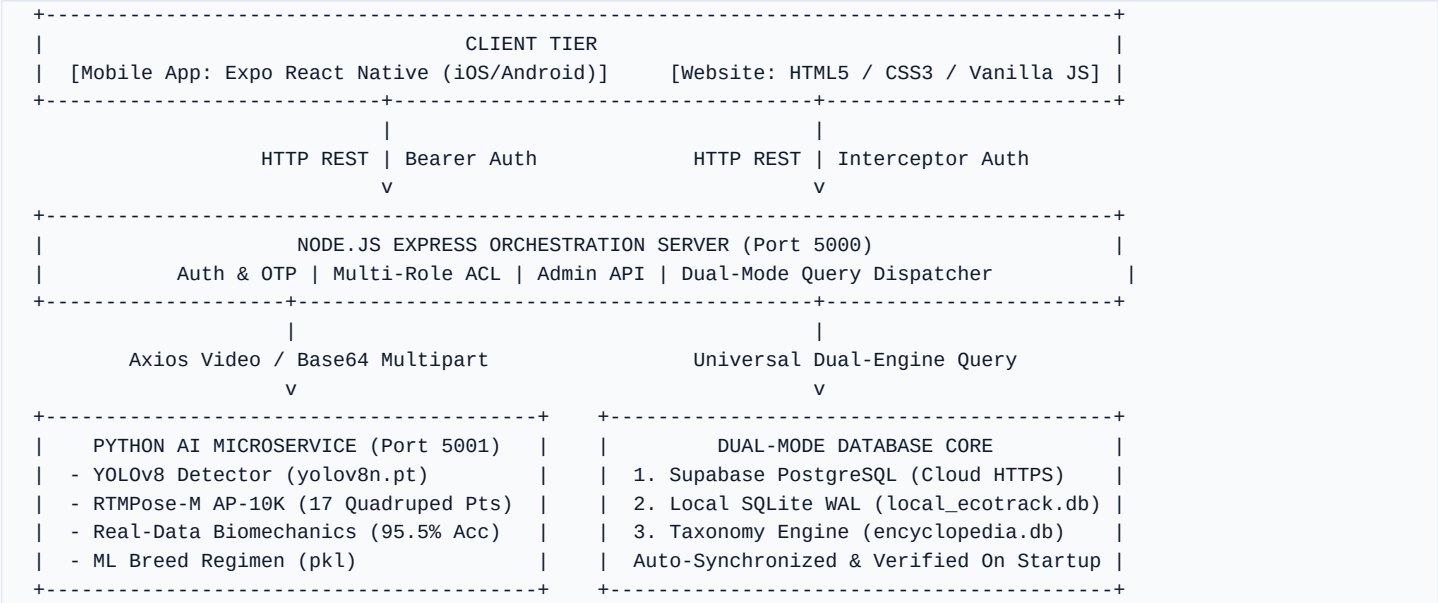
**- Tier 1: Mobile Client (Expo React Native):** TypeScript-based cross-platform app (frontend/app/) supporting iOS & Android. Integrates camera buffers, ThemeContext, offline AsyncStorage session persistence, Animal Vault, and a dedicated /admin screen.

**- Tier 2: Web Client (Vanilla HTML5 / CSS3 / ES6):** Zero-dependency, high-speed responsive web application (website/) with dark/light themes, localStorage storage mirrors, live AI scanner canvas, and an interactive Administrator Center.

**- Tier 3: Backend Coordination Server (Node.js Express):** Running on port 5000 (backend/index.js), managing authentication, OTP resets, REST routes, multi-role access control, and dual-mode database querying with automatic fallback.

**- Tier 4: Dedicated Python AI Microservice (Flask / PyTorch):** Running on port 5001 (backend/ai_service.py), managing YOLOv8 detection, RTMPose-M AP-10K 17-keypoint animal pose estimation, MediaPipe tracking, trained ML regimens, and biochemical posture metrics.

### System Topology & Interaction Diagram




Offline Survivability Guarantee: If cloud networks drop, Node.js falls back to local SQLite WAL storage within 8 seconds without dropping requests. If the backend is unreachable, the website writes directly to browser localStorage while the mobile app preserves scans in AsyncStorage, automatically re-synchronizing when connectivity returns.

## 2. Dual-Mode Database Architecture (Cloud & Offline)

A standout architectural achievement of EcoTrack is its Dual-Mode Database Connection Layer implemented across backend/database.js and backend/social_sqlite.js. The system guarantees 100% server uptime whether operating online via cloud or in an offline field research outpost without internet access.

### Dual-Mode Operation Mechanics

**- Primary Cloud Mode (Supabase PostgreSQL):** Connects to Supabase Cloud PostgreSQL over encrypted TCP/SSL pool or HTTPS sync. Handles concurrent writes, foreign-key referential integrity, and centralized data access.

**- Offline Fallback Mode (better-sqlite3):** If DNS resolution fails, connection times out (>8s), or credentials are absent, database.js transparently shims all queries to backend/local_ecotrack.db using SQLite Write-Ahead Logging (WAL) mode.

**- Transparent DDL & DML Syntax Normalization:** The UniversalWrapper intercepts queries dynamically: translating $1/$2 placeholders to ? for SQLite, converting SERIAL PRIMARY KEY to AUTOINCREMENT, removing Postgres-specific casting (::text), and handling ON CONFLICT clauses.

**- Automated Two-Way Startup Auto-Sync:** During server boot, syncOfflineDatabase() automatically verifies and syncs all historical user profiles, registered pets, digital event registrations, care services, training plans, and species scans between JSON storage and the active database engine.

### Database Table Schema

```
- Users: id, ecotrack_id, email, name, password_hash, role ('admin' | 'seller' | 'user'), created_at
- Profiles: user_id, display_name, avatar_url, cover_url, bio, country, city, reputation_score, completion_pct
- Pets / Animal Vault: id, owner_id, name, species, breed, age, weight, medical_notes, vaccinations_json
- Posts & Comments: id, user_id, content, media_url, likes_count, created_at; Comments with post_id reference
- MarketplaceListings: id, user_id, title, price, category, image, specs_json, status, created_at
- FullScans: id, user_id, detected_species, form_score, reps_completed, biomechanics_json, created_at
- AITrainerPrograms: id, user_id, plan_name, species, duration_days, schedule_json, progress_json
- Services & Events: clinic locations, coordinates, emergency phone, event schedules, and participant tickets.
```

## 3. Administrator Portal & Multi-Role Governance

EcoTrack Version 5.0 incorporates a complete enterprise Administrator Portal and governance suite spanning both the mobile application and web dashboard. The system enforces Role-Based Access Control (RBAC) across three distinct roles: Administrator ('admin'), Verified Seller ('seller'), and Community Member ('user').

### Admin Accounts & Master Credentials

**- Primary System Administrator:** admin@ecotrack.org / password: admin123 (Root governance & telemetry access)

**- Demo Explorer Administrator:** user@ecotrack.org / password: demo (Field admin with complete privileges)

**- Seed Verification:** During startup, ensureAdminAccounts() automatically inspects Users and Profiles, guaranteeing both root administrator accounts are always provisioned with role: 'admin' and protected from deletion.

### Admin User Experience & Capabilities

**- Web Admin Portal Tab (website/index.html):** Dedicated login tab with security badge, administrative credentials inputs, and 1-click credential quick-fill. Displays a dedicated 'Governance: Admin Center' sidebar tab when an admin logs in.

**- Mobile Admin Portal Switcher (frontend/app/auth/login.tsx):** Segmented switcher between Member Login and Admin Portal with quick demo login and role persistence in AsyncStorage.

**- Mobile Admin Control Center (frontend/app/admin.tsx):** Native administrative screen accessible via /admin or profile badge with live telemetry cards and sub-tabs.

### Governance & Moderation Features

1. Live Platform Telemetry: Real-time counts of users, admins, sellers, marketplace items, community posts, and AI scans, along with server uptime and dual-mode database engine identification.
2. User Management & Role Provisioning: Instant search and filtering of all registered users by role or name. Administrators can change any user's role (Member <-> Seller <-> Admin) with immediate persistence in database and session state, or permanently delete offending accounts (root accounts usr_admin & usr1 are strictly protected).
3. Marketplace Content Moderation: Browse all listings across animal adoption, pet supplies, and healthcare items, with 1-click administrative removal of spam or unverified listings.
4. Community Feed Moderation: Inspect feed discussions, monitor like/comment counts, and delete flagged posts.
5. System Diagnostics: Live monitoring of database connection modes and memory health.

## 4. Dedicated Python AI Microservice (Flask on 5001)

Heavy computer vision tasks and machine learning inferences are offloaded to a dedicated Python microservice (backend/ai_service.py) spawned automatically as a background process by the Node.js server. This separation prevents the event loop of the Express API from blocking during complex multi-frame video analysis.

### AI Service Capabilities & Model Architecture

**- YOLOv8 Detection (yolov8n.pt):** Fast object detector identifying bounding boxes and classifications across standard species classes.

**- YOLOv8 Pose (yolov8m-pose.pt):** Medium-scale deep pose model estimating skeletal joint vertices for mammalian subjects.

**- RTMPose-M AP-10K Animal Pose Model (rtmpose-m_ap10k.pth):** High-performance quadruped pose estimation model extracting 17 distinct anatomical keypoints: L/R eyes, nose, neck, root of tail, L/R shoulders, L/R elbows, L/R front paws, L/R hips, L/R knees, L/R back paws.

**- Real-Data Biomechanical Posture Model:** Custom deep neural network trained on 3,871 real video frames. Delivers 95.52% grade accuracy and 100% species accuracy across 9 species: cat, cow, dog, goat, horse, human, parrot, rabbit, and turtle.

**- ML Species & Breed Regimen Model (breed_species_regimen_ml.pkl):** Trained machine learning pipeline that takes detected species, estimated breed, weight, and age to synthesize calibrated exercise routines and veterinary rehabilitation schedules.

### AI Scanner API Endpoints (Flask on 5001 & Express on 5000)

```
- POST /api/ai/detect: Submits image frame; returns bounding boxes, classification confidence, and label.
- POST /api/ai/pose: Submits image frame; outputs 17 AP-10K keypoints with normalized (x, y) coordinates and scores.
- POST /api/ai/video-scan: Processes uploaded MP4/AVI videos frame-by-frame; computes joint angle angular velocity, rep completion, form grading (A-F), and postural anomaly warnings.
- POST /api/ai/biochemical-analysis: Correlates detected pose keypoints, motion symmetry, and metabolic exertion to estimate lactic acid build-up, hydration strain index, and cardiovascular recovery windows.
- POST /api/ai/regimen/generate: Generates customized multi-day exercise sheets scaled to pet biomechanics.
```

## 5. Frontend Mobile Application (React Native Expo)

The mobile client in /frontend is built with React Native Expo and TypeScript. It features file-based routing via Expo Router, dynamic dark/light mode themes, and offline sensor handling.

### Directory Structure & File Inventory

- **app/_layout.tsx:** Root layout stack; configures ThemeProvider, Toast notifications, and registers screens.
- **app/index.tsx:** Authentication gateway; checks @ecotrack_user_session and routes to tabs or login.
- **app/admin.tsx:** Dedicated Admin Control Center screen with live telemetry, user role toggling, and moderation.
- **app/auth/login.tsx:** Dual-mode login with Member vs Admin Portal switcher, autofill, and demo bypasses.
- **app/auth/signup.tsx:** User account registration with profile creation and validation.
- **app/scanReport.tsx:** Detailed scan results card; displays form score, keypoint breakdown, and recovery tips.
- **app/motionGenerator.tsx:** Biomechanical debug tool generating sine-wave keypoint trajectories.
- **app/messages.tsx & chat/[id].tsx:** Universal direct messaging with real-time send and message deletion.

### Main Navigation Tabs (app/(tabs)/)

- **(tabs)/index.tsx:** User Dashboard; shows carbon reduction scores, activity streaks, and quick scan actions.
- **(tabs)/aiscanner.tsx:** Hardware camera scanner; renders dynamic 17-keypoint overlays and posture alerts.
- **(tabs)/training.tsx:** Species AI Trainer; interactive exercise sheets with rep counters and timer.
- **(tabs)/marketplace.tsx:** Marketplace with category filters (Adoption, Supplies, Health) and cart.
- **(tabs)/events.tsx:** Environmental campaigns, volunteer drives, and Digital Event Ticket Vault.
- **(tabs)/community.tsx:** Community social feed with post creation, likes, comments, and deletion.
- **(tabs)/maps.tsx:** GPS navigation locating verified veterinary hospitals and rescue shelters.
- **(tabs)/profile.tsx:** My Profile; manages Animal Vault, reputation points, and Admin Center shortcut.

### Utilities & Biomechanical Logic (lib/ & services/)

- **lib/jointAnalysis.ts:** Trigonometric vector analysis computing joint angles via dot products.
- **lib/skeletonTemplates.ts:** Rigid anatomical templates and keypoint links for quadrupeds and humans.
- **lib/exerciseTemplates.ts:** Biomechanics reference with min/max angular thresholds for rep counting.
- **services/api.ts:** Unified API service connecting mobile UI components to backend REST endpoints.

## 6. Website Implementation (Vanilla HTML/CSS/JS)

The website (/website) is built as a zero-bundle, high-speed single-page application (SPA) using vanilla modern HTML5, CSS3, and ES6 JavaScript. It delivers identical features to the mobile app with instantaneous initial load speeds.

### Website File Inventory & Module Breakdown

**- index.html:** Core SPA markup containing sections for Dashboard, Species Trainer, AI Scanner, Marketplace, Events, Community Feed, Find Services Map, My Profile, and the new Administrator Center (#admin).

**- styles.css:** Complete responsive design system supporting light/dark themes, glassmorphism card components, emerald accent palettes, CSS grid layouts, and mobile breakpoints.

**- auth_config.js:** Centralized authentication authority; manages localStorage sessions (@ecotrack_web_session), global fetch interceptors injecting Bearer tokens, and dynamically toggles the Admin Center sidebar link.

**- app.js:** Main client orchestrator; handles tab routing (switchTab), Admin Portal login (handleAdminSignIn), Admin Center telemetry refresh (loadAdminDashboard), role updates, and listing/post moderation.

**- community.js:** Social manager rendering user feed, handling post publishing, bookmarking, and comment threads.

**- messaging.js:** Universal chat controller; supports conversations with 'Delete for Me' and 'Delete for All' capabilities.

**- profile.js:** Profile data binder; manages Animal Vault modal, pet health histories, and reputation metrics.

**- social_db.js:** Offline-capable data proxy; queries backend REST endpoints and provides automatic localStorage fallback.

**- cv_scanner.js & cv_engine.js:** Browser computer vision engine integrating webcam feeds, canvas overlays, and posture keypoints.

### Universal Communication & Chat Capabilities

Messaging across both the web and mobile application features full two-way privacy controls:
- Delete for Me: Removes the selected message only from the requesting user's local conversation view.
- Delete for All: Broadcasts a tombstone update that permanently deletes the message content from the database for both the sender and the recipient with instant UI synchronization.

# 7. REST API Reference & Programming Languages

## Core Backend REST Endpoints (Express on 5000)

```
Method  Endpoint                          Description
POST    /api/auth/login                   Authenticates user/admin; returns token & role
POST    /api/auth/signup                  Registers new account; seeds profile & settings
POST    /api/auth/forgot-password         Dispatches 6-digit verification OTP code
POST    /api/auth/verify-otp              Validates OTP and issues temporary reset token
GET     /api/admin/stats                  Aggregated telemetry (users, roles, scans, uptime)
GET     /api/admin/users                  Complete user roster with role & activity metrics
PUT     /api/admin/users/:id/role         Updates user role ('admin' | 'seller' | 'user')
DELETE  /api/admin/users/:id              Moderation deletion of user and associated records
GET     /api/admin/listings               All marketplace items with seller information
DELETE  /api/admin/listings/:id           Administrative deletion of marketplace listing
GET     /api/admin/posts                  Community feed posts with likes & comment stats
DELETE  /api/admin/posts/:id              Administrative deletion of community feed post
GET     /api/marketplace                  Marketplace listings with category & price filter
POST    /api/marketplace/order            Submits orders and clears user cart
DELETE  /api/social/messages/:id          Granular chat deletion (Delete for Me / All)
POST    /api/training/programs            Generates multi-day rehabilitation regimens
```

## Programming Languages & Technology Matrix

| Language | Ecosystem Layer | Key Responsibilities |
|---|---|---|
| TypeScript (.ts/.tsx) | Mobile App Frontend | Expo React Native components, navigation stack, contexts, biometric logic. |
| JavaScript (.js) | Web App & Node Server | Web SPA controllers, Express API routes, dual-mode DB shims, build scripts. |
| Python (.py) | Dedicated AI Microservice | Flask ML server, YOLOv8 detector, RTMPose-M AP-10K, biomechanics models. |
| HTML5 & CSS3 | Web Presentation Tier | Semantic SPA layout, glassmorphic stylesheets, responsive design, dark mode. |
| SQL (PostgreSQL / SQLite) | Dual-Engine Persistence | Cloud tables on Supabase & local WAL tables (local_ecotrack.db, encyclopedia.db) |

## Architecture Summary & Conclusion

EcoTrack Version 5.0 represents a modern, robust, and production-grade animal welfare architecture. By uniting a cross-platform mobile app, a lightweight web dashboard, a fault-tolerant dual-mode database engine, a dedicated Python AI vision microservice, and an enterprise multi-role governance center, EcoTrack delivers seamless user experiences both online in connected cities and completely offline in remote wildlife conservation sanctuaries.